Games and Information

Final Examination — Solutions and Grading Rubric

Prof. M. Brandt · Spring 2023 · 100 points

General grading policy. Full credit for any correct method. Consistent carry-through of an earlier arithmetic error costs at most 1 point per part. For equilibrium questions, an equilibrium stated without the supporting beliefs (on and off path) loses the belief points listed below even if the strategies are right.

Problem 1 (20 points)

(a) With observable productivity education has no value, so both types choose $e = 0$; wages are 10 for the high type and 4 for the low type.

Rubric (2 pts).

- 1 pt: $e = 0$ for both; 1 pt: wages equal productivities.

(b) In any separating PBE the low type is identified and paid 4, so she must choose $e_L = 0$ (any $e > 0$ only adds cost). Let the high type choose $e_H = e^* > 0$. Consider the beliefs

$$\mu(e) = \begin{cases} 1 & e \geq e^* \\ 0 & e < e^* \end{cases} \qquad \Longrightarrow \qquad w(e) = \begin{cases} 10 & e \geq e^* \\ 4 & e < e^*. \end{cases}$$

These are consistent with Bayes' rule on path ($\mu(0) = 0$, $\mu(e^*) = 1$).

- Low type does not mimic: $4 \geq 10 - e^* \iff e^* \geq 6$.
- High type does not switch to $e = 0$: $10 - \frac{e^*}{2} \geq 4 \iff e^* \leq 12$.
- No other deviation helps: any $e \in (0, e^*)$ earns wage 4 at positive cost, and any $e > e^*$ earns wage 10 at higher cost.

So the separating PBE are exactly: $e_L = 0$, $e_H = e^*$ with $e^* \in [6, 12]$, supported by the beliefs above. (Other off-path beliefs also work provided $w(e)$ is low enough off path; the equilibrium *outcomes* are the ones listed.)

Rubric (6 pts).

- 1 pt: $e_L = 0$ with justification.
- 1 pt: complete wage schedule / beliefs defined for every e.
- 2 pts: low type's no-mimic constraint giving $e^* \geq 6$.
- 2 pts: high type's constraint giving $e^* \leq 12$.
- Reporting only $e^* = 6$ (least-cost) as "the" separating PBE: deduct 2 pts.

(c) Suppose both types choose e_p. On path $\mu(e_p) = 0.4$ and $w(e_p) = 0.4 \cdot 10 + 0.6 \cdot 4 = 6.4$. Off path set $\mu(e) = 0$, so $w(e) = 4$ for $e \neq e_p$ (the most pessimistic beliefs make deviations least attractive, so they support the largest set of pooling equilibria). The best deviation for either type is then $e = 0$ (if $e_p > 0$).

- Low type: $6.4 - e_p \geq 4 \iff e_p \leq 2.4$.
- High type: $6.4 - \frac{e_p}{2} \geq 4 \iff e_p \leq 4.8$.

Pooling PBE: both types choose $e_p \in [0,\ 2.4]$, $w(e_p) = 6.4$, $w(e) = 4$ otherwise.

Rubric (5 pts).

- 1 pt: pooling wage 6.4.
- 1 pt: off-path beliefs specified.
- 2 pts: binding constraint is the low type's, $e_p \leq 2.4$.
- 1 pt: correct final interval $[0, 2.4]$. (Answer $[0, 4.8]$ earns 3 of 5.)

(d) Least-cost separating PBE ($e^* = 6$): high type gets $10 - 3 = 7$, low type gets 4. Pooling at $e = 0$: both types get 6.4. The low type prefers pooling; the high type prefers separating ($7 > 6.4$).

With prior π on the high type, the pooling wage is $4 + 6\pi$, while the least-cost separating payoff of the high type is still 7 (it does not depend on π). The high type prefers pooling at $e = 0$ iff $4 + 6\pi > 7$, i.e., $\pi > \frac{1}{2}$. (Then pooling at $e = 0$ Pareto-dominates the least-cost separating PBE.)

Rubric (4 pts).

- 2 pts: the four payoffs and the preference ranking.
- 2 pts: threshold $\pi > 1/2$ (1 pt if the student notes separating payoffs are independent of π but errs in algebra).

(e) *Pooling at e_p.* Consider a deviation to e'. The low type could not gain even at the best wage 10 if $10 - e' < 6.4 - e_p$, i.e., $e' > 3.6 + e_p$. The high type gains at wage 10 if $10 - \frac{e'}{2} > 6.4 - \frac{e_p}{2}$, i.e., $e' < 7.2 + e_p$. For any $e' \in (3.6 + e_p,\ 7.2 + e_p)$ the Intuitive Criterion requires $\mu(e') = 1$, and then the high type deviates. So no pooling PBE survives.

Separating with $e^* > 6$. Any $e' \in (6, e^*)$ is equilibrium-dominated for the low type ($10 - e' < 4$) but profitable for the high type at wage 10. Beliefs must put $\mu(e') = 1$, and the high type deviates. Only the least-cost (Riley) separating PBE with $e_H = 6$ survives.

Rubric (3 pts).

- 2 pts: explicit deviation interval that kills pooling.
- 1 pt: identify $e^* = 6$ as the unique survivor.

Problem 2 (20 points)

(a) Bidding one's value is weakly dominant. Revenue is the expected second-highest value: $\frac{n-1}{n+1} = \frac{2}{4} = \frac{1}{2}$.

Rubric (3 pts).

- 1 pt: truthful bidding; 2 pts: revenue 1/2.

(b) Suppose the two rivals bid $\frac{2}{3}x$. A bid $b \in [0, \frac{2}{3}]$ wins iff both rivals have $x < \frac{3b}{2}$, which has probability $\left(\frac{3b}{2}\right)^2$. The bidder maximizes

$$(v - b)\,\tfrac{9}{4}\,b^2.$$

The derivative is $\frac{9}{4}\,b\,(2v - 3b)$, which is positive for $b < \frac{2}{3}v$ and negative for $b > \frac{2}{3}v$, so $b = \frac{2}{3}v$ is the global maximizer on $[0, \frac{2}{3}]$. Bids above $\frac{2}{3}$ win for sure but cost more than bidding $\frac{2}{3}$. Hence $b(v) = \frac{2}{3}v$ is a symmetric BNE.

Revenue $= \mathbb{E}\big[\frac{2}{3}\max_i v_i\big] = \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2}$.

Rubric (6 pts).

- 1 pt: winning probability $(3b/2)^2$.
- 2 pts: FOC giving $b = 2v/3$.
- 1 pt: global optimality (sign of derivative or second-order argument, and bids above 2/3).
- 2 pts: revenue 1/2 via $\mathbb{E}[\max] = 3/4$.

(c) Revenue equivalence applies because (i) values are independent and private, (ii) bidders are risk-neutral, (iii) in the symmetric increasing equilibrium the highest-value bidder wins (same allocation as the SPA), and (iv) the lowest type $v = 0$ has expected payoff 0. Therefore every type's expected payment equals that in the SPA:

$$m(v) = \Pr(\text{win}) \cdot \mathbb{E}[\text{2nd highest} \mid \text{win}] = v^2 \cdot \tfrac{2}{3}v = \tfrac{2}{3}v^3.$$

In the all-pay auction a bidder pays her bid whether or not she wins, so $b_{AP}(v) = m(v) = \frac{2}{3}v^3$. Expected revenue is $3\,\mathbb{E}\big[\frac{2}{3}v^3\big] = 3 \cdot \frac{2}{3} \cdot \frac{1}{4} = \frac{1}{2}$.

Rubric (5 pts).

- 2 pts: conditions of the theorem (at least three of the four).
- 2 pts: $b_{AP}(v) = 2v^3/3$ (1 pt for $m(v)$ correct but not linked to the all-pay bid).
- 1 pt: revenue 1/2.

(d) Bidders still bid their values (the reserve does not affect dominance). Let $Y_1 \geq Y_2$ be the two highest values. Revenue is r if $Y_1 \geq r > Y_2$, is Y_2 if $Y_2 \geq r$, and is 0 if $Y_1 < r$.

$$\Pr(Y_1 \geq \tfrac{1}{2} > Y_2) = 3 \cdot \tfrac{1}{2} \cdot \big(\tfrac{1}{2}\big)^2 = \tfrac{3}{8}, \qquad \mathbb{E}\big[Y_2 \mathbf{1}\{Y_2 \geq \tfrac{1}{2}\}\big] = \int_{1/2}^{1} v \cdot 6v(1-v)\,dv = \tfrac{11}{32}.$$

Revenue $= \frac{1}{2} \cdot \frac{3}{8} + \frac{11}{32} = \frac{17}{32} \approx 0.531 > \frac{1}{2}$. (Equivalently, $3\int_{1/2}^{1}(2v-1)v^2\,dv = \frac{17}{32}$ using virtual values.)

Intuition: the reserve raises the price in the event that exactly one bidder values the object above $\frac{1}{2}$, which happens with probability 3/8; the cost is failing to sell when all values are below $\frac{1}{2}$ (probability 1/8), where the lost revenue is small. For uniform values the gain dominates, and $r = \frac{1}{2}$ is in fact the revenue-maximizing reserve.

Rubric (6 pts).

- 1 pt: correct description of the three revenue cases.
- 1 pt: Pr(exactly one above r) = 3/8.
- 2 pts: $\mathbb{E}[Y_2\mathbf{1}\{Y_2 \geq r\}] = 11/32$ (density of Y_2 is $6v(1-v)$).
- 1 pt: total 17/32.
- 1 pt: intuition (trade-off between extracting more from a lone high bidder and occasionally not selling).

Problem 3 (20 points)

(a) Volunteering gives $1 - c_i$ (the cleanup happens for sure). Not volunteering gives π_{-i}. Resident i volunteers iff $c_i \leq 1 - \pi_{-i}$. Because costs are independent, π_{-i} does not depend on c_i, so the best response is a cutoff rule with $c^* = 1 - \pi_{-i} = \Pr(\text{no other resident volunteers})$.

Rubric (5 pts).

- 2 pts: the two expected payoffs.
- 2 pts: comparison yields $c_i \leq 1 - \pi_{-i}$.
- 1 pt: explicit use of independence to conclude a cutoff rule.

(b) If the other resident uses cutoff c^*, she volunteers with probability $c^*/2$. The cutoff must satisfy $c^* = 1 - \frac{c^*}{2}$, so $c^* = \frac{2}{3}$. Each resident volunteers iff $c_i \leq \frac{2}{3}$ (probability $\frac{1}{3}$).

Rubric (5 pts).

- 2 pts: volunteering probability $c^*/2$.
- 2 pts: fixed-point equation.
- 1 pt: $c^* = 2/3$.

(c) With two others using cutoff c^*, $\Pr(\text{no other volunteers}) = (1 - \frac{c^*}{2})^2$. So $c^* = (1 - \frac{c^*}{2})^2$, i.e., $c^{*2} - 8c^* + 4 = 0$, with roots $4 \pm 2\sqrt{3}$. Only $c^* = 4 - 2\sqrt{3} \approx 0.536$ lies in $[0, 2]$. Each resident volunteers iff $c_i \leq 4 - 2\sqrt{3}$.

Rubric (6 pts).

- 2 pts: correct probability $(1 - c^*/2)^2$.
- 2 pts: quadratic equation.
- 2 pts: correct root selected with justification (1 pt if both roots reported).

(d) $n = 2$: $\Pr(\text{cleanup}) = 1 - \left(\frac{2}{3}\right)^2 = \frac{5}{9} \approx 0.556$.

$n = 3$: $1 - \frac{c^*}{2} = \sqrt{3} - 1$, so $\Pr(\text{nobody}) = (\sqrt{3} - 1)^3 = 6\sqrt{3} - 10 \approx 0.392$ and $\Pr(\text{cleanup}) = 11 - 6\sqrt{3} \approx 0.608$.

The probability that a given resident volunteers falls from $\frac{1}{3}$ to $2 - \sqrt{3} \approx 0.268$: each resident free-rides more when there are more potential volunteers. Here the extra resident still raises the overall probability of the cleanup, but by much less than if individual behavior were unchanged ($1 - (2/3)^3 \approx 0.704$).

Rubric (4 pts).

- 1 pt each: the two provision probabilities.
- 1 pt: individual volunteering probabilities.
- 1 pt: free-riding interpretation.

Problem 4 (20 points)

(a) Babbling: the expert sends the same message for every θ (or randomizes independently of θ); the policymaker keeps the prior after every message and chooses $a = \mathbb{E}[\theta] = \frac{1}{2}$; the expert, facing

an action that does not depend on his message, is indifferent among messages. The policymaker's expected payoff is $-\operatorname{Var}(\theta) = -\frac{1}{12} \approx -0.0833$.

Rubric (4 pts).

- 2 pts: strategies and beliefs; 2 pts: payoff $-1/12$.

(b) The expert sends "low" if $\theta < x_1$ and "high" otherwise. The policymaker's best responses are the conditional means $a_1 = \frac{x_1}{2}$ and $a_2 = \frac{1+x_1}{2}$. The expert's ideal point is $\theta + b$, so the type $\theta = x_1$ must be indifferent: $x_1 + b = \frac{a_1+a_2}{2} = \frac{2x_1+1}{4}$, which gives

$$x_1 = \tfrac{1}{2} - 2b = \tfrac{3}{10}, \qquad a_1 = \tfrac{3}{20} = 0.15, \qquad a_2 = \tfrac{13}{20} = 0.65.$$

Since quadratic loss is single-peaked, types below x_1 strictly prefer a_1 and types above prefer a_2, so the messages are incentive compatible. Such an equilibrium exists iff $x_1 > 0$, i.e., $b < \frac{1}{4}$.

Rubric (6 pts).

- 1 pt: conditional-mean actions.
- 2 pts: indifference condition at x_1 (must include the bias b).
- 2 pts: $x_1 = 3/10$ and both actions.
- 1 pt: existence condition $b < 1/4$.

(c) On an interval of length ℓ the loss is the variance of a uniform, $\ell^2/12$, times the probability ℓ. So

$$\mathbb{E}[u_R] = -\frac{x_1^3 + (1-x_1)^3}{12} = -\frac{0.027 + 0.343}{12} = -\frac{37}{1200} \approx -0.0308,$$

compared with $-\frac{1}{12} \approx -0.0833$ under babbling. Informative communication cuts the policymaker's expected loss by almost two thirds.

Rubric (4 pts).

- 2 pts: correct formula; 1 pt: value $-37/1200$; 1 pt: comparison.

(d) Let $0 = x_0 < x_1 < \cdots < x_N = 1$ and $a_k = \frac{x_{k-1}+x_k}{2}$. Indifference of type x_k between a_k and a_{k+1}:

$$x_k + b = \frac{a_k + a_{k+1}}{2} = \frac{x_{k-1} + 2x_k + x_{k+1}}{4} \implies (x_{k+1} - x_k) = (x_k - x_{k-1}) + 4b.$$

So interval lengths are $\ell_k = \ell_1 + 4b(k-1)$, and $\sum_{k=1}^{N} \ell_k = N\ell_1 + 2bN(N-1) = 1$ requires $\ell_1 = \frac{1-2bN(N-1)}{N} > 0$, i.e., $2bN(N-1) < 1$.

For $b = \frac{1}{50}$: $N(N-1) < 25$, so the largest N is 5 ($5 \cdot 4 = 20 < 25$ but $6 \cdot 5 = 30 > 25$). Then $\ell_1 = \frac{1-0.8}{5} = 0.04$, the lengths are $0.04,\ 0.12,\ 0.20,\ 0.28,\ 0.36$, and the boundaries are

$$0,\ 0.04,\ 0.16,\ 0.36,\ 0.64,\ 1.$$

Rubric (6 pts).

- 2 pts: derivation of the $4b$ increment.
- 1 pt: existence condition $2bN(N-1) < 1$.
- 1 pt: $N = 5$.
- 2 pts: correct boundaries (1 pt if lengths correct but boundaries mis-summed).

Problem 5 (20 points)

(a) A strategy for Player 1 is a pair (action in X, action in Y). Player 1's best responses by state:

- If Player 2 plays L: U in state X ($4 > 1$) and D in state Y ($2 > 0$), i.e., UD.
- If Player 2 plays R: D in state X ($2 > 0$) and U in state Y ($3 > 1$), i.e., DU.

Player 2 against UD: L yields $\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 3 = \frac{5}{2}$ and R yields 0, so L is a best response. Against DU: L yields 0 and R yields $\frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 1 = \frac{5}{2}$, so R is a best response. The pure BNE are

$$(UD,\, L) \qquad \text{and} \qquad (DU,\, R).$$

Rubric (6 pts).

- 1 pt: Player 1's strategies are type-contingent.
- 2 pts: Player 1's best responses in each state.
- 2 pts: Player 2's expected payoff comparisons.
- 1 pt: both equilibria listed and no others. (One equilibrium only: 4 of 6.)

(b) In a common-value auction each bidder has a noisy estimate of the same unknown value. Winning means one's estimate was the highest, so conditional on winning the estimate is biased upward; a bidder who bids based on the unconditional estimate overpays on average. Rational bidders avoid the curse by bidding on $\mathbb{E}[V \mid \text{own signal is the highest}]$, i.e., shading more. With private values each bidder knows her own value, and winning reveals nothing about it, so there is no curse.

Rubric (4 pts).

- 2 pts: selection effect of winning; 1 pt: rational bidders condition on winning; 1 pt: private-value contrast.

(c) The first-price auction yields weakly higher (generically strictly higher) revenue. In the second-price auction truthful bidding remains weakly dominant regardless of risk attitude, so revenue is unchanged. In the first-price auction, raising one's bid trades a lower surplus if winning for a higher chance of winning; a risk-averse bidder values this "insurance" against losing and bids more aggressively than a risk-neutral one, which raises revenue.

Rubric (4 pts).

- 1 pt: FPA higher; 1 pt: SPA unaffected; 2 pts: insurance intuition.

(d) (i) Two bidders, $r = \frac{1}{2}$: the price is r when exactly one value exceeds $\frac{1}{2}$ (probability $2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$) and the second value when both exceed it. Revenue $= \frac{1}{2} \cdot \frac{1}{2} + \int_{1/2}^{1} v \cdot 2(1-v)\, dv = \frac{1}{4} + \frac{1}{6} = \frac{5}{12} \approx 0.417$.

(ii) Three bidders, no reserve: revenue $= \frac{3-1}{3+1} = \frac{1}{2}$.

Option (ii) is better ($\frac{1}{2} > \frac{5}{12}$). This is an instance of the Bulow–Klemperer result: one additional bidder is worth more to the seller than the optimal reserve price.

Rubric (6 pts).

- 3 pts: revenue 5/12 (1 pt for the case split, 2 pts for the computation).
- 2 pts: revenue 1/2.
- 1 pt: correct comparison.